CNENGR202 Instrumentation and Control

Part 1 L2 – Mathematical models

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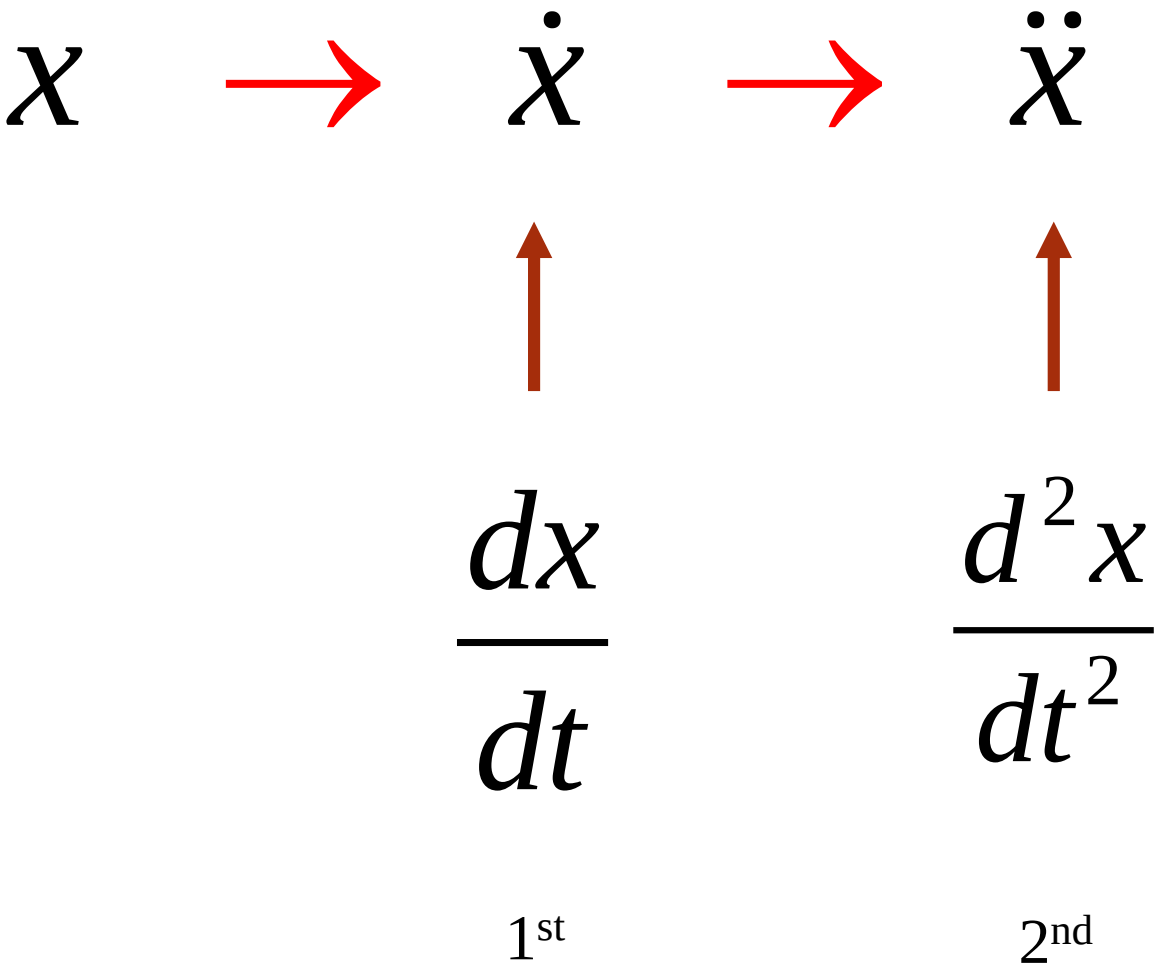
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Outline

- Introduction
- Mechanistic models
- Electronic models
- Generalised 1st and 2nd order systems

Introduction - 1

- Standard notation



Introduction - 2

- **Black Box Models** 以尽可能简单的方式将输入和输出变量联系起来，而不考虑内部机制
  - Relate the input and output variables in the simplest possible manner, without regard to internal mechanisms
  - e.g. Transfer Function with parameters estimated from data using graphical or statistical methods

- **Mechanistic and Electronic Models**
  - The internal structure of the model is important, i.e. each element will have physical significance
  - Usually obey known physical laws as perceived by the model builder / scientist

Introduction - 3

- Newton's Law for Translational Motion

  - F is the sum of all the forces applied to a body (Newtons or N), m is mass of the body (kg) and a is the acceleration (m/s^2)

  - In SI units, a force of 1N will impart an acceleration of $1m/s^2$ to a mass of 1kg

$$F = ma$$

Introduction – 4

- Ohm's law

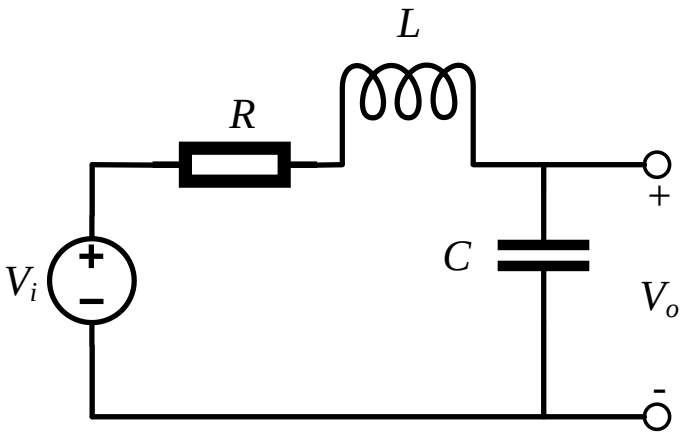
  - The current i through a conductor is directly proportional to the potential difference V across the two points

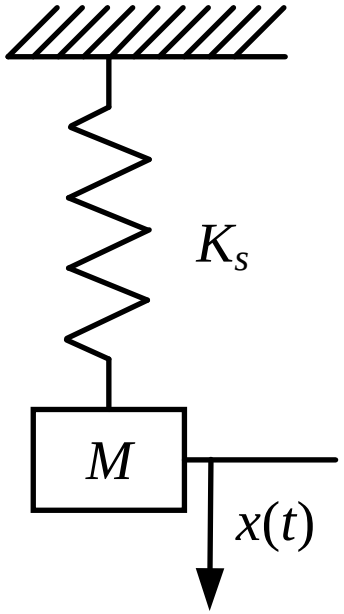
  - Constant of proportionality R (resistance)

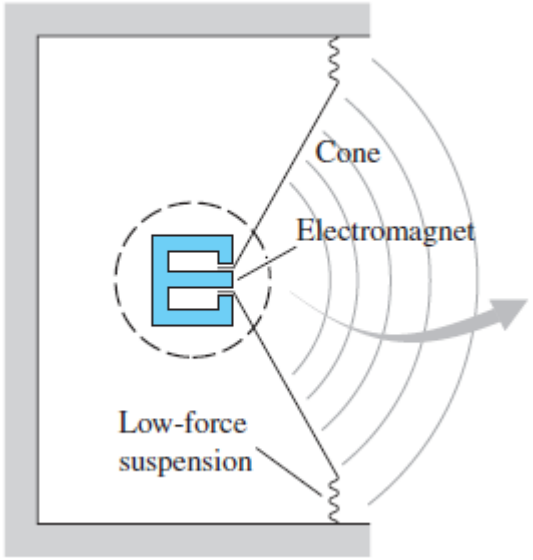
  - A conductor has a resistance of 1 ohm if on application of a potential difference of 1 volt, a current of 1 ampere flows through it

$$V = Ri$$

Introduction - 5





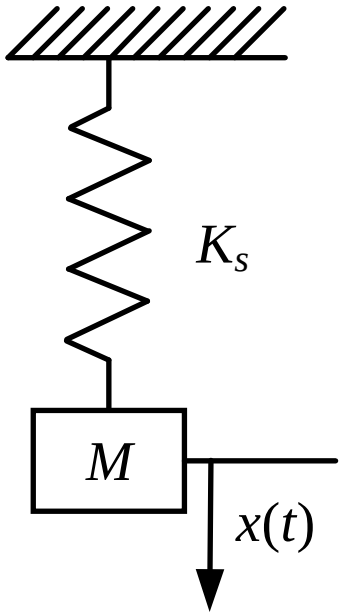


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Mechanistic Models - 1

- Harmonic Oscillator
- Assumptions
  - Constant mass
  - No air resistance
  - Spring has no mass
  - No internal friction in the spring
  - Spring force proportional to displacement



$$F = K_s\, x(t)$$

Where: K_s is spring constant
x is the displacement

Mechanistic Models - 2

- Newton's second law states that force equals mass times acceleration

$$M\frac{d^2x(t)}{dt^2}$$

- The force acting on the system is generated by the spring

$$K_s x(t)$$

- The force is opposing the acceleration (equation of simple harmonic motion)

$$M\frac{d^2x(t)}{dt^2} = -K_s x(t)$$

- Rearranging

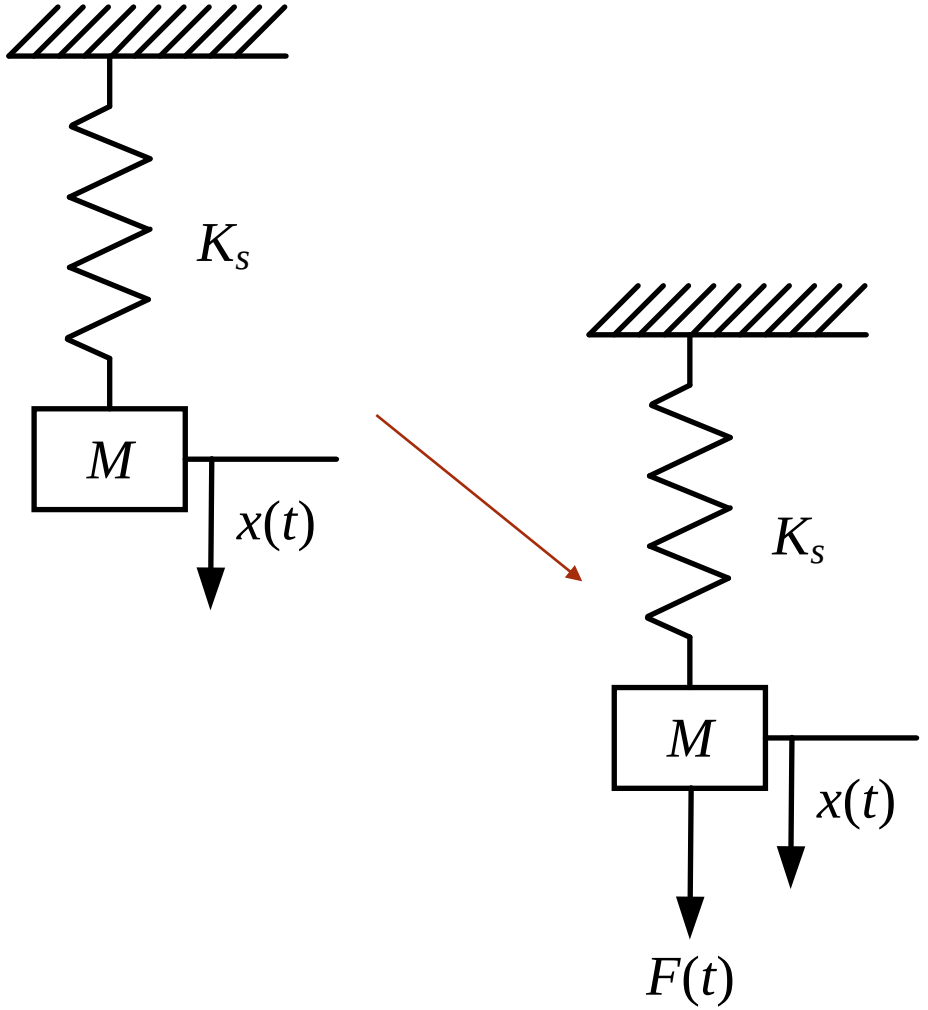
$$M\frac{d^2x(t)}{dt^2} + K_s x(t) = 0$$

Mechanistic Models - 3

- Harmonic Oscillator with External Force

$$M\frac{d^2x}{dt^2} = -K_s x + F(t)$$

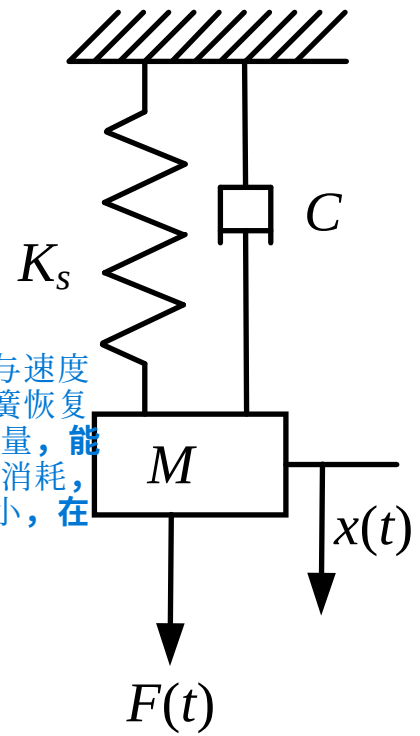
$$M\frac{d^2x}{dt^2} + K_s x = F(t)$$



Mechanistic Models - 4

- Mass–Spring–Damper (MSD) System

- Additional Assumption
  - Damping force proportional to rate of change of displacement



阻尼力始终与运动方向相反，一直做负功，而弹簧的恢复力有时与速度同向有时则相反，**在无阻尼（等幅运动）情况下**，一个周期内弹簧恢复**力是不做功的** 理想弹簧不存在能量衰减，能够不断储存和释放能量，**能**量大小保持不变。 阻尼是弹簧在能量释放和吸收过程中存在能量消耗，比如弹簧分子间内能的增加等，弹簧在振动过程中的振幅不断减小，**在无其他外界激励的情况下振幅无限接近于零。**

$$C\frac{dx}{dt}$$

Where: C is damping coefficient

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+K_s x=F(t)$$

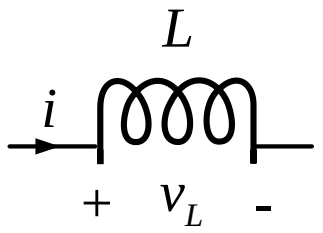
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Electronic Models - 1

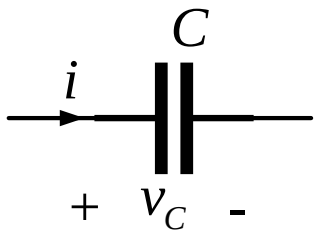
- Kirchhoff's voltage law states that the sum of voltage drops in a loop must equal to zero
- Potential difference across a inductor

$$v_L = L\frac{di}{dt}$$



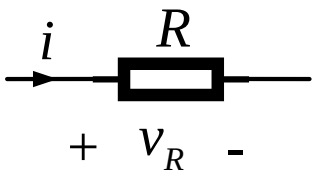
- Current across a capacitor

$$i = C\frac{dv_C}{dt}$$



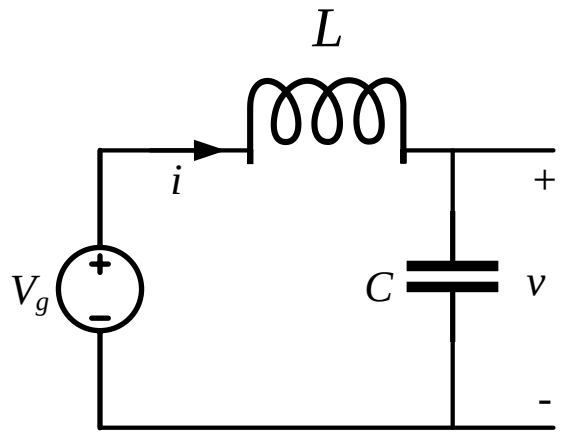
- Potential difference across resistor

$$v_R = R \cdot i$$



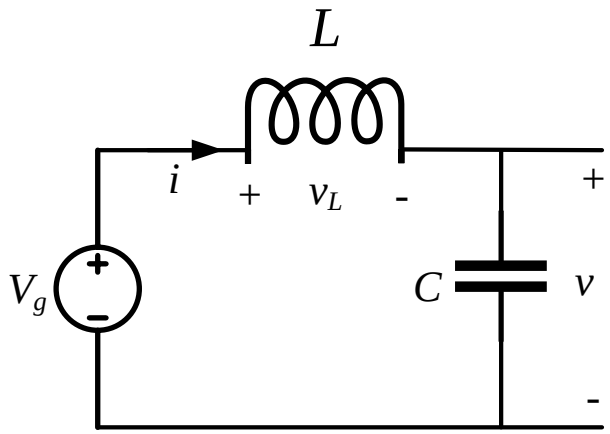
Electronic Models - 2

- *LC* Circuit
  - Assumptions
    - No resistance in L or C
    - Energy is not dissipated in L or C
    - Small voltage and current



Electronic Models – 3

$$i = C\frac{dv}{dt} \quad \rightarrow \quad \frac{di}{dt} = C\frac{d^2v}{dt^2}$$



$$L\frac{di}{dt} + v = V_g \quad \rightarrow \quad LC\frac{d^2v}{dt^2} + v = V_g$$

v_L

Outline

- Introduction
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Generalised Forms - 1

- Generalised 1st Order Model

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+K_sx=F(t)$$

- M: 0
- C: τ
- K_s: 1
- $F(t)$: $Ku(t)$

- Output: x
- Input or forcing function: $u(t)$
- Time constant: τ
- Steady state gain: K

Why?

$$\tau\frac{dx}{dt}+x=Ku(t)$$

Generalised Forms - 2

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+K_sx=F(t)$$

- Generalised 2nd Order Model
  - Output: x
  - Input or forcing function: $u(t)$
  - Damping ratio: ζ
  - Natural frequency: ω_n
  - Steady state gain: K/ω_n^2

- M: 1
- C: $2\zeta\omega_n$
- K_s: ω_n^2
- $F(t)$: $Ku(t)$

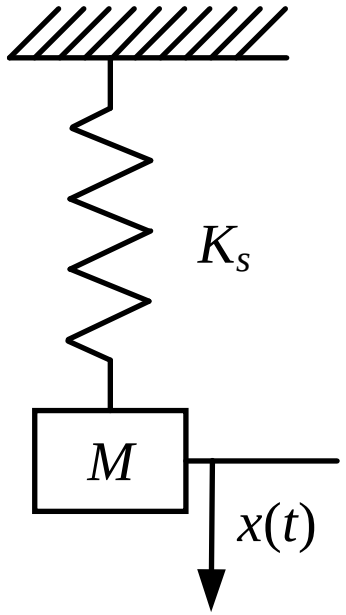
$$\frac{d^2x}{dt^2}+2\zeta\omega_n\frac{dx}{dt}+\omega_n^2x=Ku(t)$$

Why?

Generalised Forms - 3

- Harmonic Oscillator

$$M\frac{d^2x}{dt^2}+K_s x=0$$



Generalised Forms - 4

Spring constant

$$M\frac{d^2x}{dt^2}+K_s x=0$$

$$\frac{d^2x}{dt^2}+2\zeta\omega_n\frac{dx}{dt}+\omega_n^2 x=Ku(t)$$

$$\frac{d^2x}{dt^2}+\frac{K_s}{M}x=0$$

$$x = distance$$

$$u=0$$

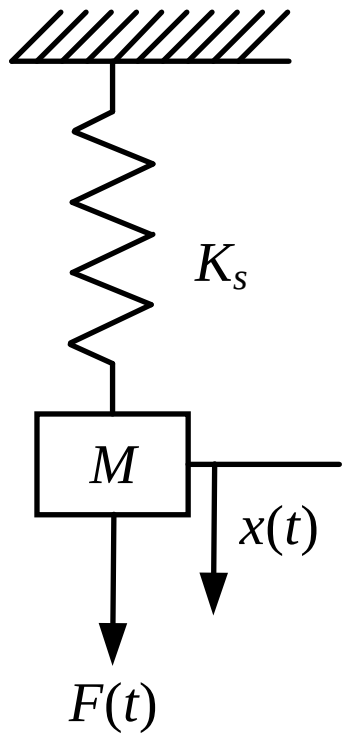
$$\omega_n^2=K_s/M$$

$$2\zeta\omega_n=0\rightarrow\zeta=0$$

Generalised Forms - 5

- Harmonic Oscillator with external force

$$M\frac{d^2x}{dt^2}+K_s x=F(t)$$



Generalised Forms - 6

$$M\frac{d^2x}{dt^2}+K_sx=F(t)$$

Spring constant

$$\frac{d^2x}{dt^2}+2\zeta\omega_n\frac{dx}{dt}+\omega_n^2x=Ku(t)$$

Steady State Gain

Generalised Forms - 7

$$M\frac{d^2x}{dt^2}+K_s x=F(t)$$

$$\frac{d^2x}{dt^2}+\frac{K_s}{M}x=\frac{1}{M}F(t)$$

$$\frac{d^2x}{dt^2}+\omega_n^2 x=Ku(t)$$

$x = distance$

$u=F(t)$

$K=1/M$

$\omega_n^2=K_s/M$

$2\zeta\omega_n=0\rightarrow\zeta=0$

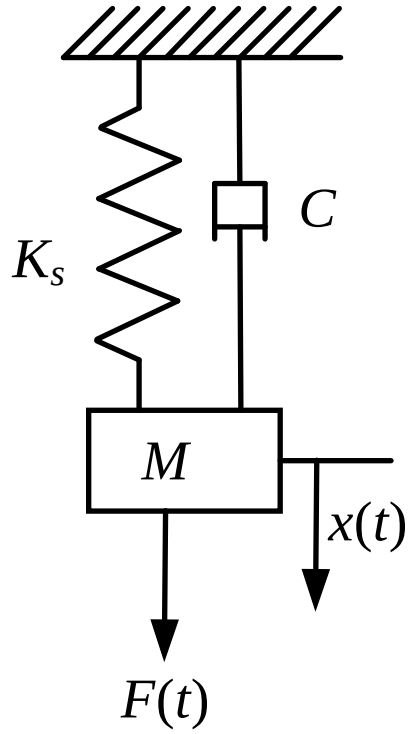
Damping ratio = zero

What does this mean?

Generalised Forms - 8

- Mass–Spring–Damper (MSD) System

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+K_s x=F(t)$$



Generalised Forms - 9

$$M\frac{d^2x}{dt^2}+C\frac{dx}{dt}+K_s x=F(t)$$

$$\frac{d^2x}{dt^2}+\frac{C}{M}\frac{dx}{dt}+\frac{K_s}{M}x=\frac{1}{M}F(t)$$

$$\frac{d^2x}{dt^2}+2\zeta\omega_n\frac{dx}{dt}+\omega_n^2 x=Ku(t)$$

$x = distance$

$u=F(t)$

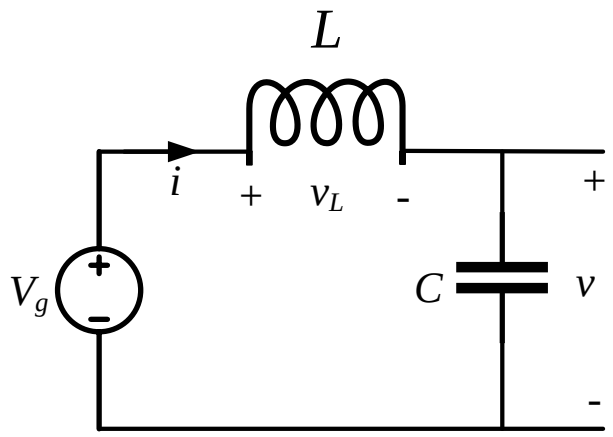
$K=1/M$

$\omega_n^2=K_s/M$

$2\zeta\omega_n=C/M$

Generalised Forms - 10

- *LC* Circuit



$$LC\frac{d^2v}{dt^2}+v=V_g$$

Generalised Forms - 11

$$LC\frac{d^2v}{dt^2}+v=V_g$$

$$\frac{d^2v}{dt^2}+\frac{1}{LC}v=\frac{1}{LC}V_g$$

$$\frac{d^2x}{dt^2}+2\zeta\omega_n\frac{dx}{dt}+\omega_n^2x=Ku(t)$$

$$x=v$$

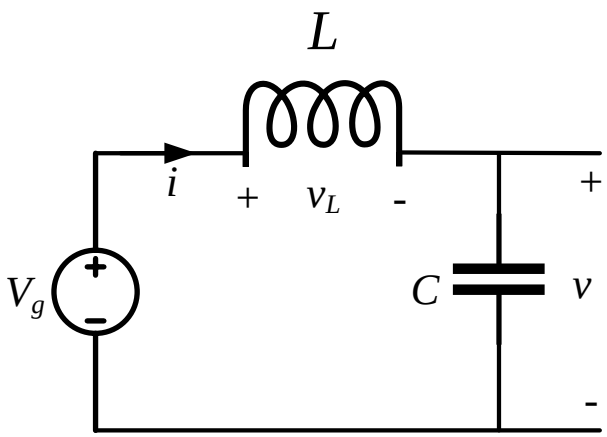
$$u=V_g$$

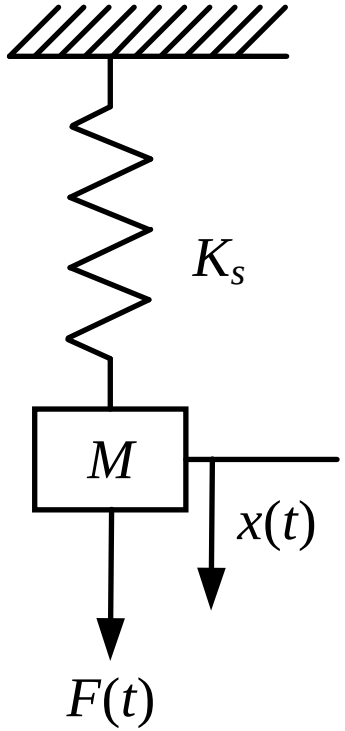
$$K=1/LC$$

$$\omega_n^2=1/LC$$

$$2\zeta\omega_n=0\rightarrow\zeta=0$$

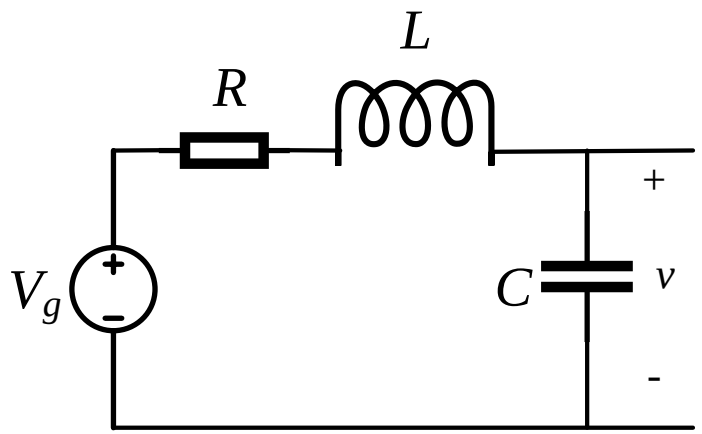
Generalised Forms - 12

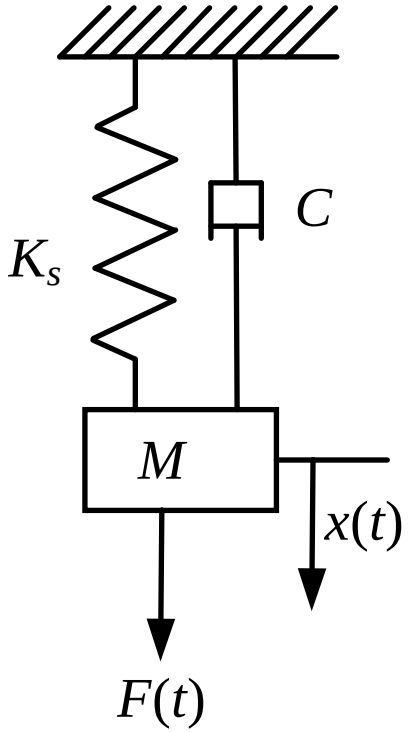




$$\frac{d^2x}{dt^2} + \omega_n^2 x = Ku(t)$$

Generalised Forms - 13

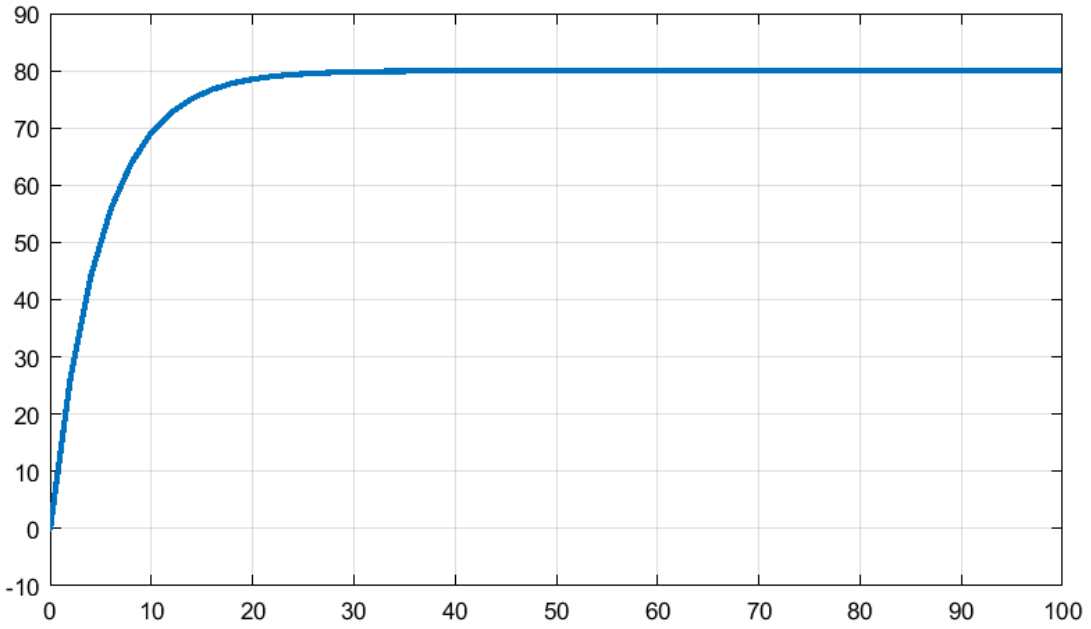




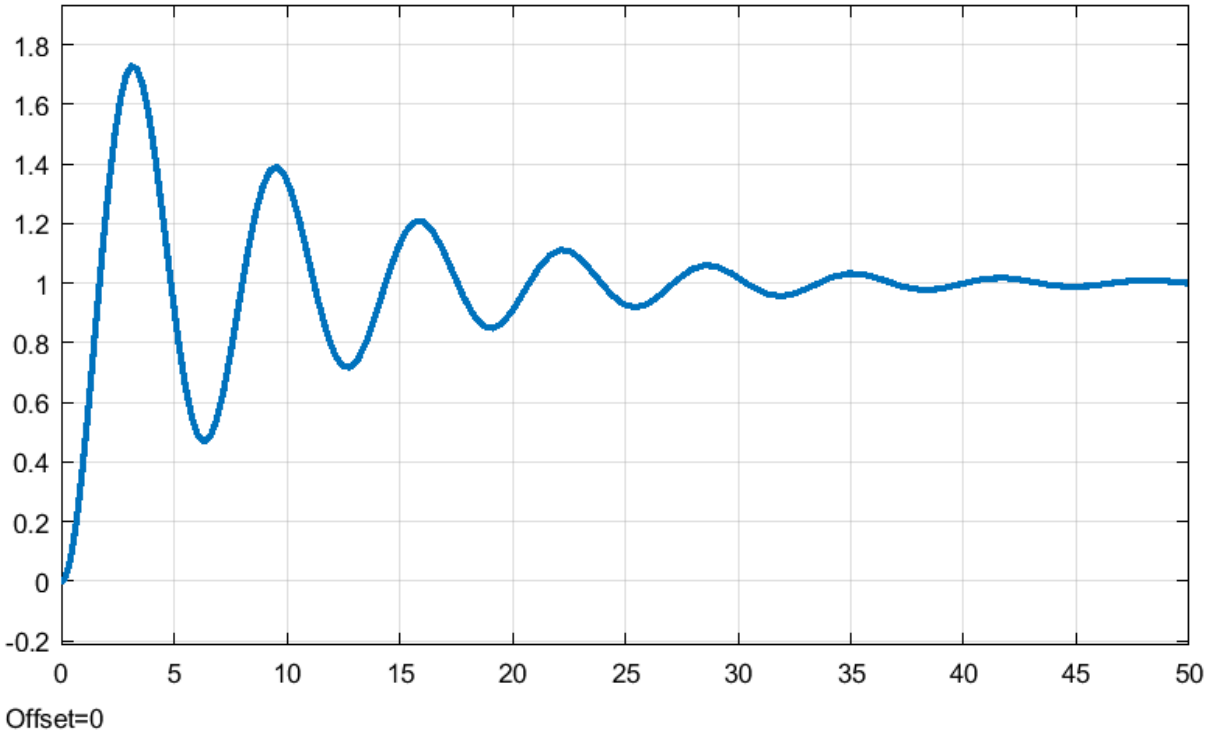
$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t)$$

Generalised Forms - 14

$$\tau \frac{dx}{dt} + x = Ku(t)$$



$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t)$$



Home question

Write the dynamic equation for the right figure.

